Math 242, S09
Exam 3

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	30	
2	25	
3	25	
4	20	
Total	100	

Good Luck !

Problem 1. (30 points) Find the Laplace or inverse Laplace transforms of the following functions:

(a) $\mathcal{L}\{t^2 - 3\cosh 2t\}$

$= \mathcal{L}\{t^2\} - 3\mathcal{L}\{\cosh 2t\}$

$= \frac{2!}{s^3} - 3 \cdot \frac{s}{s^2 - 2^2}$

$= \frac{2}{s^3} - \frac{3s}{s^2-4}$

(b) $\mathcal{L}\{t^3 e^{4t}\}$ (Hint: $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$ where $F(s) = \mathcal{L}\{f(t)\}$.)

$= \mathcal{L}\{t^3\}(s-4)$

$= \frac{3!}{(s-4)^4}$

$= \frac{6}{(s-4)^4}$

(c) $\mathcal{L}^{-1}\Big\{\frac{1}{s(s^2+9)}\Big\}$ (Hint: $\mathcal{L}^{-1}\big(\frac{F(s)}{s}\big) = \int_0^t \mathcal{L}^{-1}\{F(s)\}d\tau$.)

$= \int_0^t \mathcal{L}^{-1}\{\frac{1}{s^2+9}\}\,d\tau$

$= \int_0^t \frac{1}{3}\sin 3\tau \,d\tau$

$= -\frac{1}{9}\cos 3\tau \Big|_{\tau=0}^{t}$

$= -\frac{1}{9}\cos 3t + \frac{1}{9}$

Problem 2. (25 points) Use Laplace transform to solve the initial value problem:

$$x'' - 9x = 0; \quad x(0) = 5, \; x'(0) = 3.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(t)$, then

$$\mathcal{L}\{x''(t)\} = s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0)$$
$$= s^2 X(s) - 5s - 3$$

Then, $\mathcal{L}\{x'' - 9x\} = \mathcal{L}\{0\}$ gives us

$$\mathcal{L}\{x''\} - 9\mathcal{L}\{x\} = 0$$
$$[s^2 X(s) - 5s - 3] - 9X(s) = 0$$
$$(s^2 - 9)X(s) = 5s + 3$$
$$X(s) = \frac{5s+3}{s^2-9}$$

Thus,

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{5s+3}{s^2-9}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{5s}{s^2-9}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{s^2-9}\right\}$$
$$= 5\cosh 3t + \sinh 3t$$

Problem 3. (25 points) Use Laplace transform to solve the initial value problem

$$x'' + 4x' - 5x = 2; \quad x(0) = x'(0) = 0.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x'(t)\} = s X(s)$$

$$\mathcal{L}\{x''(t)\} = s^2 X(s)$$

Then, $\mathcal{L}\{x''+4x'-5x\} = \mathcal{L}\{2\}$ gives us

$$\mathcal{L}\{x''\} + 4\mathcal{L}\{x'\} - 5\mathcal{L}\{x\} = \frac{2}{s}$$

$$s^2 X(s) + 4s X(s) - 5X(s) = \frac{2}{s}$$

$$(s^2+4s-5) X(s) = \frac{2}{s}$$

$$X(s) = \frac{2}{s(s^2+4s-5)}$$

$$\frac{2}{s(s^2+4s-5)} = \frac{2}{s(s-1)(s+5)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+5}$$

$$= \frac{A(s-1)(s+5)+Bs(s+5)+Cs(s-1)}{s(s-1)(s+5)}$$

let $s=0$, we have $A\cdot(-1)\cdot 5 = 2 \Rightarrow A = -\frac{2}{5}$

let $s=1$, we have $B\cdot 1\cdot 6 = 2 \Rightarrow B = \frac{1}{3}$

let $s=-5$, we have $C\cdot(-5)\cdot(-6) = 2 \Rightarrow C = \frac{1}{15}$

Thus,

$$x(t) = \mathcal{L}^{-1}\left\{\frac{2}{s(s^2+4s-5)}\right\} = \mathcal{L}^{-1}\left\{\frac{-2/5}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1/3}{s-1}\right\} + \left\{\frac{1/15}{s+5}\right\}$$

$$= -\frac{2}{5} + \frac{1}{3}e^{t} + \frac{1}{15}e^{-5t}$$

Problem 4. (20 points)
(a) Let $f(t) = t$ and $g(t) = e^{2t}$, compute the convolution

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\, d\tau.$$

Solution: $(f*g)(t) = t * e^{2t} = \int_0^t \tau \cdot e^{2(t-\tau)} d\tau$

$$= e^{2t}\int_0^t \tau e^{-2\tau} d\tau$$

$$= e^{2t}\left[\tau \cdot \left(\frac{e^{-2\tau}}{(-2)}\right)\Big|_{\tau=0}^{t} + \int_0^t \frac{e^{-2\tau}}{2} d\tau\right]$$

$$= e^{2t}\left[-\frac{te^{-2t}}{2} + \left(\frac{e^{-2\tau}}{-4}\right)\Big|_{\tau=0}^{t}\right]$$

$$= e^{2t}\left[-\frac{te^{-2t}}{2} - \frac{e^{-2t}}{4} + \frac{1}{4}\right]$$

$$= -\frac{t}{2} - \frac{1}{4} + \frac{e^{2t}}{4}$$

(b) **Theorem of Differential Tansforms** tells us that $\mathcal{L}\{-tf(t)\} = F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$. Use this formula to compute $\mathcal{L}\{t\cos 4t\}$.

Solution: $\mathcal{L}\{t\cos 4t\} = -\mathcal{L}\{-t\cos 4t\}$

$$= -\frac{d}{ds}\left[\mathcal{L}\{\cos 4t\}\right]$$

$$= -\frac{d}{ds}\left[\frac{s}{s^2+16}\right]$$

$$= -\left[\frac{1\cdot(s^2+16) - s\cdot(2s)}{(s^2+16)^2}\right]$$

$$= -\frac{(-s^2+16)}{(s^2+16)^2}$$

$$= \frac{s^2-16}{(s^2+16)^2}$$